

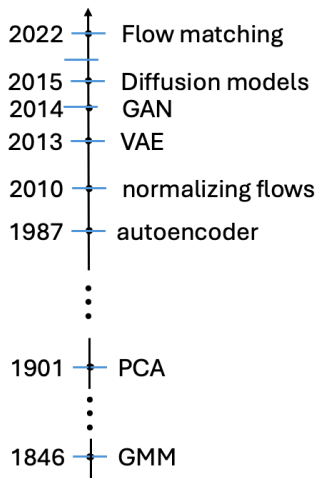
COMM114: Generative AI - Diffusion Models (part 1)

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Diffusion models

Diffusion Models was invented in 2015.



Images generated by diffusion models



pictures from "Improved Distribution Matching Distillation for Fast Image Synthesis"

Video generated by diffusion models



Check video via the [youtube link](#).

Recap from flow matching model

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $\mathbf{X}_1$ be $p_{data}(\mathbf{x})$ (i.e., $p_{\mathbf{X}_1}(\mathbf{x}_1) = p_{data}(\mathbf{x})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Let

$$\mathbf{u}_t^{\text{target}}(\mathbf{X}_t) = \frac{d\mathbf{X}_t}{dt} = \frac{d\beta_t}{dt} \mathbf{X}_0 + \frac{d\alpha_t}{dt} \mathbf{X}_1 \quad [ODE]$$

Suppose $\mathbf{u}_t^{\text{target}}(\mathbf{X}_t)$ is defined as above.

- It can be shown that $\mathbf{X}_t$, the solution to the ODE, transforms the distribution p_{init} at $t = 0$ to the data distribution p_{data} at $t = 1$.
- In other words, the probability path of $\mathbf{X}_t$ satisfy the boundary probability conditions.

Recap from flow matching model (cont.)

Objective of an FM model:

- To train a neural network $\mathbf{u}_t^\theta(\cdot)$ by regression with respect to $\mathbf{u}_t^{\text{target}}(\mathbf{X}_t)$:

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \mathbf{u}_t^{\text{target}}(\mathbf{X}_t) \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \frac{d}{dt} \mathbf{X}_t \right\|^2 \\ &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{u}_t^\theta(\mathbf{X}_t) - \frac{d\beta_t}{dt} \mathbf{X}_0 - \frac{d\alpha_t}{dt} \mathbf{X}_1 \right\|^2 \end{aligned}$$

- Once the neural network $\mathbf{u}_t^\theta(\cdot)$ is well trained, one can generate **new data samples** by solving the following ODE:

$$\mathbf{u}_t^\theta(\mathbf{X}_t) = \frac{d\mathbf{X}_t}{dt} \text{ subject to } \mathbf{X}_0 \sim p_{\text{init}}(\cdot) \quad [ODE]$$

Objective of a diffusion model

- While a flow matching model trains a neural network $\mathbf{u}_t^\theta(\cdot)$ by regression with respect to the velocity field $\mathbf{u}_t^{\text{target}}(\mathbf{X}_t)$, a **diffusion model** attempts to train a neural network by regression with respect to a so-called **score function**.
- Once a diffusion model is well trained, new data samples can be generated by simulating an ODE which is different but equivalent to the one in flow matching.

Prior Knowledge

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $\mathbf{X}_1$ be $p_{data}(\mathbf{x})$ (i.e., $p_{\mathbf{X}_1}(\mathbf{x}_1) = p_{data}(\mathbf{x})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- We can obtain two equivalent ODEs

$$\boxed{\mathbf{u}_t^{\text{target}}(\mathbf{X}_t) = \frac{d\mathbf{X}_t}{dt} = \frac{d\beta_t}{dt} \mathbf{X}_0 + \frac{d\alpha_t}{dt} \mathbf{X}_1} \quad [ODE] \quad (1)$$

$\Updownarrow$

$$\boxed{\frac{d\mathbf{X}_t}{dt} = \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) + \frac{d \log \alpha_t}{dt} \mathbf{X}_t} \quad [ODE] \quad (2)$$

where $p_{t|1}(\mathbf{X}_{t|1} | \mathbf{X}_1)$ is the conditional probability distribution of $\mathbf{X}_t$ given $\mathbf{X}_1$

Exercise

What is the conditional distribution of $\mathbf{X}_t$ given $\mathbf{X}_1$?

Prior Knowledge (cont.)

We show ODE (2) is equivalent to ODE (1):

1. The conditional distribution of $\mathbf{X}_t$ given $\mathbf{X}_1$ is a Gaussian distribution:

$$p_{t|1}(\mathbf{x}_t|\tilde{\mathbf{x}}) = \mathcal{N}(\alpha_t \tilde{\mathbf{x}}, \beta_t^2 \mathbf{I}_d) \quad \text{where } \tilde{\mathbf{x}} \text{ drawn from } p_{data}(\cdot)$$

2. The **score function** $\nabla \log p_{t|1}(\mathbf{x}_t|\tilde{\mathbf{x}})$ takes the form of

$$\boxed{\nabla_{\mathbf{x}_t} \log p_{t|1}(\mathbf{x}_t|\tilde{\mathbf{x}}) = -\frac{\mathbf{x}_t - \alpha_t \tilde{\mathbf{x}}}{\beta_t^2}} \quad (3)$$

- Derivation of (3):

$$\begin{aligned} & \nabla_{\mathbf{x}_t} \log p_{t|1}(\mathbf{x}_t|\tilde{\mathbf{x}}) \\ &= \nabla_{\mathbf{x}_t} \log \left(\frac{1}{(2\pi)^{\frac{d}{2}} \beta_t^d} e^{-\frac{1}{2\beta_t^2} \|\mathbf{x}_t - \alpha_t \tilde{\mathbf{x}}\|^2} \right) \\ &= -\nabla_{\mathbf{x}_t} \left(\frac{1}{2\beta_t^2} \|\mathbf{x}_t - \alpha_t \tilde{\mathbf{x}}\|^2 + \text{constant} \right) \\ &= -\frac{\mathbf{x}_t - \alpha_t \tilde{\mathbf{x}}}{\beta_t^2} \end{aligned}$$

Prior Knowledge (cont.)

1. From (3), $p_{t|1}(\mathbf{X}_{t|1}|\mathbf{X}_1)$ takes the form of

$$\boxed{\nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) = -\frac{\mathbf{X}_t - \alpha_t \mathbf{X}_1}{\beta_t^2}} \quad (4)$$

2. Plugging (4) into (2) produces

$$\begin{aligned} \frac{d\mathbf{X}_t}{dt} &= \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t|\mathbf{X}_1) + \frac{d \log \alpha_t}{dt} \mathbf{X}_t \\ \Leftrightarrow \frac{d\mathbf{X}_t}{dt} &= \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \frac{\alpha_t \mathbf{X}_1 - \mathbf{X}_t}{\beta_t^2} + \frac{d \log \alpha_t}{dt} \mathbf{X}_t \\ \Leftrightarrow \frac{d\mathbf{X}_t}{dt} &= \frac{d \log \alpha_t}{dt} (\alpha_t \mathbf{X}_1 - \mathbf{X}_t) - \frac{d\beta_t}{dt} \frac{\alpha_t \mathbf{X}_1 - \mathbf{X}_t}{\beta_t} + \frac{d \log \alpha_t}{dt} \mathbf{X}_t \\ \Leftrightarrow \frac{d\mathbf{X}_t}{dt} &= \frac{d \log \alpha_t}{dt} \alpha_t \mathbf{X}_1 - \frac{d\beta_t}{dt} \frac{\alpha_t \mathbf{X}_1 - \mathbf{X}_t}{\beta_t} \quad [\text{use } \mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1] \\ \Leftrightarrow \frac{d\mathbf{X}_t}{dt} &= \frac{d\alpha_t}{dt} \mathbf{X}_1 + \frac{d\beta_t}{dt} \mathbf{X}_0 \quad [\text{proof is complete}] \end{aligned}$$

Prior Knowledge (cont.)

- Let PDF of $\mathbf{X}_0$ be $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ (i.e., $p_{\mathbf{X}_0}(\mathbf{x}_0) = p_{init}(\mathbf{x})$)
- Let PDF of $\mathbf{X}_1$ be $p_{data}(\mathbf{x})$ (i.e., $p_{\mathbf{X}_1}(\mathbf{x}_1) = p_{data}(\mathbf{x})$)
- Define $\mathbf{X}_t = \beta_t \mathbf{X}_0 + \alpha_t \mathbf{X}_1$ where α_t and β_t satisfy:
 - α_t and β_t are scalar differentiable functions over $t \in [0, 1]$.
 - When $t = 0$, $\alpha_{t=0} = 0$ and $\beta_{t=0} = 1$.
 - When $t = 1$, $\alpha_{t=1} = 1$ and $\beta_{t=1} = 0$.
- Consider

$$\boxed{\frac{d\mathbf{X}_t}{dt} = \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d \beta_t^2}{dt} \right) \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) + \frac{d \log \alpha_t}{dt} \mathbf{X}_t} \quad [ODE]$$

- It can be shown that $\mathbf{X}_t$, the solution to the above ODE, transforms the distribution $p_{init} = \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ at $t = 0$ to the data distribution p_{data} at $t = 1$.
- In other words, the probability path of $\mathbf{X}_t$ satisfy the boundary probability conditions.

Objective of a diffusion model

- **Input:** data samples $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$.
- **Assumption:** Assume $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the data distribution $p_{data}(\cdot)$.
- We train a neural network $\mathbf{s}_t^\theta(\mathbf{X}_t)$ over $t \in [0, 1]$ from the dataset $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$ by regression with respect to $\nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1)$.
- Objective function for $\mathbf{s}_t^\theta(\cdot)$:

$$\begin{aligned}
 & \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{s}_t^\theta(\mathbf{X}_t) - \nabla_{\mathbf{X}_t} \log p_{t|1}(\mathbf{X}_t | \mathbf{X}_1) \right\|^2 \\
 &= \arg \min_{\theta} \mathbb{E}_{t, \mathbf{X}_1, \mathbf{X}_0} \left\| \mathbf{s}_t^\theta(\mathbf{X}_t) + \frac{\mathbf{X}_t - \alpha_t \mathbf{X}_1}{\beta_t^2} \right\|^2 \\
 &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \mathbf{x}_t \sim p_{t|1}(\cdot | \mathbf{x})} \left\| \mathbf{s}_t^\theta(\mathbf{x}_t) + \frac{\mathbf{x}_t - \alpha_t \mathbf{x}}{\beta_t^2} \right\|^2 \\
 &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left\| \mathbf{s}_t^\theta(\alpha_t \mathbf{x} + \beta_t \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\beta_t} \right\|^2 \\
 &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\frac{1}{\beta_t^2} \left\| \beta_t \mathbf{s}_t^\theta(\alpha_t \mathbf{x} + \beta_t \boldsymbol{\epsilon}) + \boldsymbol{\epsilon} \right\|^2 \right] \quad (5)
 \end{aligned}$$

Objective of a diffusion model (cont.)

Issues with the the objective function (5):

- In practice, it is found that the neural network model $s_t^\theta(x_t)$ trained via (5) is not good enough. The trained model cannot produce realistic data samples.
- This is because when β_t approaches zero, the objective function is too large. On the other hand, when β_t approaches 1, the objective function is relatively small. The trained model does not predict uniformly well over $t \in [0, 1]$.

Objective of a diffusion model (cont.)

Alternative objective function:

- **Input:** data samples $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$.
- **Assumption:** Assume $\{\mathbf{x}_i\}_{i=1}^n$ are drawn from the data distribution $p_{data}(\cdot)$.
- We train a neural network $\hat{\epsilon}_t^\theta(\mathbf{X}_t)$ over $t \in [0, 1]$ from the dataset $\{\mathbf{x}_i \in \mathbb{R}^d\}_{i=1}^n$ by regression with respect to $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
- Objective function for $\hat{\epsilon}_t^\theta(\cdot)$:

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\mathbf{x}_t) - \epsilon \right\|^2 \right] \\ &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{x} \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\alpha_t \mathbf{x} + \beta_t \epsilon) - \epsilon \right\|^2 \right] \end{aligned} \quad (6)$$

- The associated ODE in terms of $\hat{\epsilon}_t^\theta(\cdot)$ is:

$$\begin{aligned} & \boxed{\frac{d\mathbf{x}_t}{dt} = \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \frac{-(\mathbf{x}_t - \alpha_t \tilde{\mathbf{x}})}{\beta_t^2} + \frac{d \log \alpha_t}{dt} \mathbf{x}_t \quad \text{where} \quad \tilde{\mathbf{x}} \sim p_{data}} \\ & \approx \boxed{d\mathbf{x}_t = \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d\beta_t^2}{dt} \right) \frac{-\hat{\epsilon}_t^\theta(\mathbf{x}_t)}{\beta_t} + \frac{d \log \alpha_t}{d} t \mathbf{x}_t} \quad \text{where} \quad \mathbf{x}_0 \sim p_{init} \end{aligned}$$

Objective of a diffusion model (cont.)

The associated ODE in terms of $\hat{\epsilon}_t^\theta(\cdot)$ is:

$$\begin{aligned} dx_t &= \left(\beta_t^2 \frac{d \log \alpha_t}{dt} - \frac{1}{2} \frac{d \beta_t^2}{dt} \right) \frac{-\hat{\epsilon}_t^\theta(x_t)}{\beta_t} + \frac{d \log \alpha_t}{d} t x_t \text{ where } x_0 \sim p_{init} \\ \Leftrightarrow dx_t &= \left(\beta_t^2 d \log \alpha_t - \frac{1}{2} d \beta_t^2 \right) \frac{-\hat{\epsilon}_t^\theta(x_t)}{\beta_t} + d \log \alpha_t x_t \text{ where } x_0 \sim p_{init} \\ \Leftrightarrow dx_t &= (d \beta_t - \beta_t d \log \alpha_t) \hat{\epsilon}_t^\theta(x_t) + d \log \alpha_t x_t \text{ where } x_0 \sim p_{init} \end{aligned} \quad (7)$$

Remark: Once the neural network $\hat{\epsilon}_t^\theta(x_t)$ is well trained, one can generate new data samples by simulating the ODE (7).

Training procedure

Algorithm 1 Training Procedure for a diffusion model

- 1: Input: A dataset of samples drawn from p_{data} , neural network $\hat{\epsilon}_t^\theta$
- 2: **while** not converge **do**
- 3: Set $\mathcal{L}(\theta) = 0$
- 4: **for** each mini-batch of data **do**
- 5: Select a sample x from the dataset
- 6: Sample a random time $t \sim \mathcal{U}[0, 1]$
- 7: Sample noise $\epsilon \sim \mathcal{N}(0, \mathbf{I}_d)$
- 8: Compute loss

$$\mathcal{L}(\theta) = \mathcal{L}(\theta) + \left\| \epsilon_t^\theta(x_t) - \epsilon \right\|^2$$

- 9: **end for**
- 10: Update the model parameter via gradient descent

$$\theta \leftarrow \theta - \eta \frac{d\mathcal{L}(\theta)}{d\theta}$$

- 11: **end while**
-

Sampling from a diffusion model

- Given the ODE $d\mathbf{x}_t = (d\beta_t - \beta_t d\log \alpha_t) \hat{\epsilon}_t^\theta(\mathbf{x}_t) + d\log \alpha_t \mathbf{x}_t$ where $\mathbf{x}_0 \sim p_{init}$, applying the Euler method produces

$$\mathbf{x}_{t+h} - \mathbf{x}_t = \left(\beta_{t+h} - \beta_t - \beta_t \frac{\alpha_{t+h} - \alpha_t}{\alpha_t} \right) \hat{\epsilon}_t^\theta(\mathbf{x}_t) + \frac{\alpha_{t+h} - \alpha_t}{\alpha_t} \mathbf{x}_t \text{ where } \mathbf{x}_0 \sim p_{init}$$

$$\Leftrightarrow \mathbf{x}_{t+h} = \left(\beta_{t+h} - \beta_t \frac{\alpha_{t+h}}{\alpha_t} \right) \hat{\epsilon}_t^\theta(\mathbf{x}_t) + \frac{\alpha_{t+h}}{\alpha_t} \mathbf{x}_t \text{ where } \mathbf{x}_0 \sim p_{init}$$

$$\Leftrightarrow \mathbf{x}_{t+h} = \beta_{t+h} \hat{\epsilon}_t^\theta(\mathbf{x}_t) + \alpha_{t+h} \frac{(\mathbf{x}_t - \beta_t \hat{\epsilon}_t^\theta)}{\alpha_t} \text{ where } \mathbf{x}_0 \sim p_{init}$$

Remark: The term $\frac{(\mathbf{x}_t - \beta_t \hat{\epsilon}_t^\theta)}{\alpha_t}$ is in fact an approximation of the clean data sample.

Remark: The above sampling method is alternatively referred to as DDIM sampling procedure. Please check the [paper link](#)

Sampling from a diffusion model (cont.)

Algorithm 2 Sampling from a diffusion model

- 1: Input: neural network $\hat{\epsilon}_t^\theta$, number of steps k
 - 2: Set $t = 0$
 - 3: Set stepsize $h = \frac{1}{k}$
 - 4: Draw a sample $\mathbf{x}_0 \sim p_{init} = \mathcal{N}(0, \mathbf{I})$
 - 5: **for** $i = 1, \dots, k$ **do**
 - 6: $\mathbf{x}_{t+h} = \beta_{t+h} \hat{\epsilon}_t^\theta(\mathbf{x}_t) + \alpha_{t+h} \frac{(\mathbf{x}_t - \beta_t \hat{\epsilon}_t^\theta)}{\alpha_t}$ [DDIM sampling procedure]
 - 7: Set $t \leftarrow t + h$
 - 8: **end for**
 - 9: return $\mathbf{x}_{t=1}$
-

Remark 1: In principle, any ODE solver can be applied to perform sampling for a diffusion model.

Conventional Diffusion Formulation

- In a conventional diffusion model, we let PDF of the diffusion state $\mathbf{z}_0$ at timestep $t = 0$ follows the data distribution $p_{data}(\cdot)$.
- A forward diffusion process $\{\mathbf{z}_t | T \geq t \geq 0\}$ is defined as

$$\mathbf{z}_t = \alpha_t \mathbf{z}_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{z}_0 \sim p_{data}(\cdot)$$

- α_t monotonically decreases as t increases until T .
- σ_t monotonically increases as t increases until T .
- One then train a neural network $\hat{\epsilon}_t^\theta(\mathbf{z}_t)$ to predict the noise ϵ for $T \geq t \geq 0$:

$$\begin{aligned} & \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{z}_0 \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\mathbf{z}_t) - \epsilon \right\|^2 \right] \\ &= \arg \min_{\theta} \mathbb{E}_{t \sim \mathcal{U}[0,1], \mathbf{z}_0 \sim p_{data}, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)} \left[\left\| \hat{\epsilon}_t^\theta(\alpha_t \mathbf{z}_0 + \sigma_t \epsilon) - \epsilon \right\|^2 \right] \end{aligned} \quad (8)$$

Conventional Diffusion Formulation (cont.)

- Once the DNN model is trained, one can do the sampling by solving the following ODE:

$$dz_t = \overbrace{\left[\frac{d \log \alpha_t}{dt} z_t - \frac{1}{2} \left(\frac{d \sigma_t^2}{dt} - 2 \frac{d \log \alpha_t}{dt} \sigma_t^2 \right) \nabla_z \log p_{t|0}(z_t; \alpha_t, \sigma_t) \right]}^{d(z_t, t)} dt \quad z_T \in \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

where $p_{t|0}(z_t; \alpha_t, \sigma_t)$ is the conditional distribution of z_t given z_0 .

- $\nabla_z \log p_{t|0}(z_t; \alpha_t, \sigma_t)$ can be approximated to be

$$\begin{aligned} \nabla_z \log p_{t|0}(z_t; \alpha_t, \sigma_t) &= \frac{-(z_t - \alpha_t z_0)}{\sigma_t^2} \\ &\approx \frac{-\sigma_t \hat{\epsilon}_t^\theta(z_t)}{\sigma_t^2} \\ &= \frac{-\hat{\epsilon}_t^\theta(z_t)}{\sigma_t} \end{aligned}$$

Conventional Diffusion Formulation (cont.)

- The ODE can be approximated as

$$\begin{aligned} dz_t &\approx \overbrace{\left[\frac{d \log \alpha_t}{dt} z_t + \frac{1}{2} \left(\frac{d \sigma_t^2}{dt} - 2 \frac{d \log \alpha_t}{dt} \sigma_t^2 \right) \frac{\hat{\epsilon}_t^\theta(z_t)}{\sigma_t} \right]}^{d(z_t, t)} dt \quad z_T \in \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \\ &= \left[z_t d \log \alpha_t + \frac{1}{2} (d \sigma_t^2 - 2 \sigma_t^2 d \log \alpha_t) \frac{\hat{\epsilon}_t^\theta(z_t)}{\sigma_t} \right] \quad z_T \in \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \\ &= \left[z_t d \log \alpha_t + (d \sigma_t - \sigma_t d \log \alpha_t) \hat{\epsilon}_t^\theta(z_t) \right] \quad z_T \in \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \end{aligned}$$

Sampling from a conventional diffusion model

- Given the ODE $dz_t = \left[z_t d \log \alpha_t + (d\sigma_t - \sigma_t d \log \alpha_t) \hat{\epsilon}_t^\theta(z_t) \right]$ where $z_T \in \mathcal{N}(0, \sigma_T^2 \mathbf{I})$, applying the Euler method produces

$$z_{t-h} - z_t = \left(\sigma_{t-h} - \sigma_t - \sigma_t \frac{\alpha_{t-h} - \alpha_t}{\alpha_t} \right) \hat{\epsilon}_t^\theta(z_t) + \frac{\alpha_{t-h} - \alpha_t}{\alpha_t} z_t$$

$$\Leftrightarrow z_{t-h} = \left(\sigma_{t-h} - \sigma_t \frac{\alpha_{t-h}}{\alpha_t} \right) \hat{\epsilon}_t^\theta(z_t) + \frac{\alpha_{t-h}}{\alpha_t} z_t$$

$$\Leftrightarrow z_{t-h} = \sigma_{t-h} \hat{\epsilon}_t^\theta(z_t) + \alpha_{t-h} \frac{(z_t - \sigma_t \hat{\epsilon}_t^\theta)}{\alpha_t}$$

Remark: The term $\frac{(z_t - \sigma_t \hat{\epsilon}_t^\theta)}{\alpha_t}$ is in fact an approximation of the clean data sample.

Remark: The above sampling method is alternatively referred to as DDIM sampling procedure. Please check the [paper](#) [link](#)

References

- Lecture notes by MIT
- Tutorial on Flow Matching for Generative Modeling

Questions?